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A review on cost allocation methods in collaborative transportation

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Abstract

Collaboration in transportation between two or more agents is becoming an important approach to find efficient solutions or plans. Efficiency can be measured in, for example, lower cost or more flexibility. An important aspect of the collaboration is to decide on how to share the benefits—for example, cost, profit, or resources. There are many sharing mechanisms or cost allocations proposed in the literature. Some are based on simple proportional rules and others are based on theoretical concepts found in game theory. We provide a survey on cost allocation methods found in the literature on collaborative transportation, including problems on planning, vehicle routing, traveling salesman, distribution, and inventory. A total of 55 scientific articles compose the main part of the survey, most of them published between 2010 and 2015. We identify more than 40 cost allocation methods used in this stream of literature. We describe the theoretical basis for the main methods as well as the cases where they are used. We also report savings from the collaborations when they are based on industrial data. Some directions for future research are discussed.

Keywords: transportation; cost allocation; collaborative logistics; game theory; group decisions and negotiations

1. Introduction

Collaboration among different agents is an effective way to improve logistic operations. One example is that two companies use a common inventory location to share fixed costs. Another is that two companies that distribute one or several products to the same region hire a third-party logistic provider to distribute for both companies. A third is two transport companies sharing information on transportation orders to enable return trips or backhauling. There are, of course, many more possibilities for collaboration. One key question in all collaboration is how to share the costs or profits and another is how to build and manage the coalition. In this article, we focus on the first question.

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A simple approach for cost allocation is to use a proportional allocation that can be based on the overall volume or weight of the products transported. A more advanced approach is to use principles based on game theory. Modern game theory began with the idea regarding the existence of mixed-strategy equilibria in two-person zero-sum games and its proof by John von Neumann. One of the fundamental books considering cooperative games of several players by John von Neumann and Oskar Morgenstern was published in 1944 and entitled Theory of Games and Economic Behavior (von Neumann and Morgenstern, 1944). Different methods based on game theory can be shown to have different theoretical properties. A frequently used cost allocation method is the so-called Shapley values (Shapley, 1953). This is a popular approach in many applications and was rewarded, among other achievements, when L. S. Shapley received the Nobel Memorial Price in Economic Sciences 2012 together with A. E. Roth.

During the last 10 years, the number of articles on collaborative logistics has increased considerably. Borm et al. (2001), Curiel (2008), Fiestras-Janeiro et al. (2011), and Audy et al. (2012b) review related literature on cost allocation, but in a broader context not completely focused on transportation. On the other hand, Cruijssen et al. (2007) and Pérez-Bernabeu et al. (2015) review literature on collaborative transportation, but with only brief mention to cost allocation. Our review concentrates on work connecting both areas, which constitute a stream of literature actively developed in recent years. More specifically, we address the areas of transportation planning, vehicle routing, traveling salesman, distribution, and inventory. There are many different application areas among these and even more different cost allocation methods proposed.

The articles reviewed in this survey have been gathered in three stages. In the first stage, we proceeded with a structured search on the web page of the Institute for Scientific Information (2015). As topic search we used "cost allocation" together with "transportation," "traveling," or "routing." In the second stage, we did a careful selection of articles among the results of this search. Our selection was aimed at identifying articles where transportation plays an important role in the cost computation and the cost must be allocated among different members. As a result of this search, we identified 35 articles. In the third stage, this list was complemented with other articles that we came across naturally while conducting the review, either because they were frequently cited in the previously selected articles or by tracking references to them. In this way we added 20 more articles, so the selected list for this review included 55 articles. Among these, 36 have been published between 2010 and 2015. The main contribution of this article is to provide a survey of cost allocation methods encountered in the literature of collaborative transportation, as well as to characterize the contributions in this literature. We also provide a condensed mathematical description of the main approaches used in the literature.

The outline of the paper is as follows. In Section 2, we classify all 55 articles according to five type of problems on collaborative transportation. In Section 3, we summarize some basics on cooperative game theory and outline the cost allocation methods used in the literature on collaborative transportation. In Section 4, we present other characteristics encountered in the articles including data features, sector of application and findings on methods performance. We conclude with some remarks and a future research agenda in Section 5.

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2 Problem classification

We have classified the articles on collaborative transportation according to five main types of problems: transportation planning, traveling salesman, vehicle routing, joint distribution, and inventory related problems. Table 1 summarizes this classification (note some articles may belong to more than one group). Most of these have been published between 2010 and 2015, as depicted in Fig. 1.

The selected articles cover a broad range of journals in the operational research, transportation, economics, and other research fields, as shown in Table 2.

In the following, we give a brief account of all the articles.

2.1. Transportation planning

In the well-known transportation problem (Tolstoi, 1930; Hitchcock, 1941), a certain good is available in limited quantities at several origins and must be transported to destinations at minimum total cost such that the demand quantities at the destinations are satisfied and the supply availability at the origins is respected. The optimal solution to this problem can be found by a linear programming model where the variables are the flows from each origin to each destination. Different agents can collaborate in their transportation plans, for example, by allowing one agent's origin point to supply another agent's demand point. Several variations to this problem have been studied in the literature or a similar structure appears as part of a larger planning problem involving other decisions. Samet et al. (1984) tackle the problem of finding a cost allocation to each destination in the transportation

Table 1
Problem classification and references

Problem	Articles
Transportation planning	Samet et al. (1984), Sakawa et al. (2001), Frisk et al. (2010), Massol and Tchung-Ming (2010), Lehoux et al. (2011), Audy et al. (2012a), Flisberg et al. (2015), Guajardo and Rönnqvist (2015), Guajardo et al. (2015), Wang et al. (2015).
Traveling salesman	Fishburn and Pollak (1983), Dror (1990), Potters et al. (1992), Derks and Kuipers (1997), Engevall et al. (1998), Faigle et al. (1998), Engevall et al. (2004), Bläser and Ram (2008), Estévez-Fernández et al. (2009), Berger and Bierwirth (2010), Caprara and Letchford (2010), Yengin (2012), Toriello and Uhan (2013), Sun et al. (2015).
Vehicle routing	Göthe-Lundgren et al. (1996), Engevall et al. (2004), Krajewska and Kopfer (2006), Krajewska et al. (2008), Caprara and Letchford (2010), Cruijssen et al. (2010), Liu et al. (2010), Dahl and Derigs (2011), Özener (2014), Vanovermeire and Sörensen (2014b), Zakharov and Shchegryaev (2015).
Joint distribution	Hamers et al. (1999), Özener and Ergun (2008), Agarwal and Ergun (2010), Audy et al. (2011), Granot et al. (2011), Dai and Chen (2012), Yilmaz and Savasaneril (2012), Lozano et al. (2013), Nguyen et al. (2014), Vanovermeire and Sörensen (2014a), Vanovermeire and Sörensen (2014b), Vanovermeire et al. (2014), Dai and Chen (2015), Hezarkhani et al. (2015), Yang et al. (2015).
Inventory	Anupindi et al. (2001), Anily and Haviv (2007), Wong et al. (2007), Fiestras-Janeiro et al. (2012), Cheng et al. (2013), Fiestras-Janeiro et al. (2013), Özener et al. (2013), Fang and Cho (2014).

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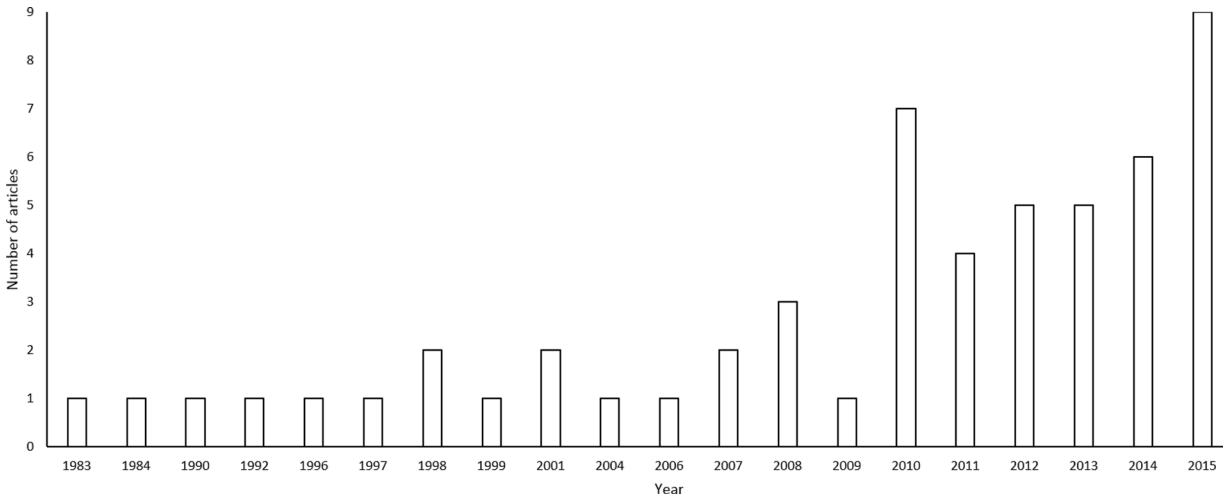

Fig. 1. Number of articles per year of publication.

plan obtained from optimizing the classic linear programming formulation. Sakawa et al. (2001) deal with a problem combining transportation planning with production planning in a housing material manufacturer with multiple production and sales bases in Japan. Once an optimal plan is found, they deal with the problem of how to allocate costs to each base. Massol and Tchung-Ming (2010) study cooperative strategies in the transportation of liquified natural gas among different supplier countries. The transportation problem in this case is formulated as a "maximum" profit instead of "minimum" cost, and the allocation problem consists of splitting the gains of the collaborative plan among different country members. Lehoux et al. (2011) also present a max profit problem, where a pulp and paper producer can collaborate with one of his merchants by different contractual agreements. Here, the transportation costs are charged by the number of trucks used in delivering products rather than on the flow quantities. Frisk et al. (2010) address a case study of eight forestry companies in Sweden. Wood bartering and backhauling can be used in a collaborative transportation plan where supply meets demand of these companies at minimum cost. After a collaborative optimal plan is found, the authors deal with the problem of how much cost should be allocated to each company. An indication of the increasing interest in collaboration is that this paper received the "Best EJOR Paper on Innovative Applications of OR" 2012. A followup to this case is presented in Audy et al. (2012a), where different business models and leadership behavior are studied regarding the allocation problem. Also, Guajardo and Rönnqvist (2015) study the problem of which coalition should form and how much cost is allocated to each company in a cooperative setting restricted by a cardinality bound. This problem relates to what in cooperative game theory is known as coalition structure, namely, a partition of the set of players. Guajardo et al. (2015) study constructive and blocking power concepts in this case study for both the cost allocation and the coalition structure problems. Flisberg et al. (2015) perform a country-wide study accounting for all transport of forest fuel in Sweden, according to a transportation planning problem. The optimal collaborative plan is used to estimate the potential savings of a coalition involving 29 company members. After the optimal plan has been found, the problem of how to allocate the cost among

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Table 2
Number of articles per journal

Journal	No. of articles
European Journal of Operational Research	8
Operations Research	5
OR Spectrum	4
Transportation Research Part E	4
International Journal of Game Theory	3
Journal of the Operational Research Society	3
Mathematical Programming	3
Computers & Industrial Engineering	2
Mathematical Problems in Engineering	2
Transportation Science	2
American Mathematical Monthly	1
Annals of Operations Research	1
Applied Mathematics and Computation	1
Automation and Remote Control	1
Decision Support Systems	1
Energy	1
Energy Economics	1
European Journal of Industrial Engineering	1
Games and Economic Behavior	1
International Journal of Logistics-Research and Applications	1
International Journal of Production Economics	1
International Journal of Shipping and Transport Logistics	1
International Transactions in Operational Research	1
Journal of Service Science and Management	1
Mathematics of Operations Research	1
Manufacturing & Service Operations Management	1
Operations Research Letters	1
Theory of Computing Systems	1
TOP	1
Total	55

the 29 members is addressed. In Wang et al. (2015), several distribution centers can collaborate to satisfy the demand from different units.

2.2. *Traveling salesman*

Given an origin or home city and a set of other cities, the widely studied traveling salesman problem (TSP) consists of finding a minimum distance tour that starts at the origin, visits each city exactly once and finishes at the origin (see, e.g., a review by Laporte, 1992a). A cost allocation problem linked to the TSP consists of splitting the cost of the tour among the visited cities. A pioneer reference is Fishburn and Pollak (1983), where the cost allocation problem is motivated by an academic who needs to split the cost of a multistop trip sponsored by different host institutions. Dror (1990) studies

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the cost allocation in the TSP distinguishing the case with and without a fixed home city. Potters et al. (1992) formally introduces the cost allocation in the TSP as "traveling salesman games," and defines problems with a fixed route and without a fixed route. When the route is fixed, the cost of any subset of cities is computed respecting the order in which the cities are visited in the original tour. When the route is not fixed, the cost of a subset of cities is computed by finding the optimal tour involving these cities. Later references on the fixed-route problem include Derks and Kuipers (1997), Yengin (2012), and Sun et al. (2015), and on the version without fixed routes Faigle et al. (1998) and Caprara and Letchford (2010). Cost allocation in several variations of the TSP has also been studied. Bläser and Ram (2008) study the case where the cost matrix satisfies the triangle inequality and relate the cost allocation in the TSP to cycle cover games. Estévez-Fernández et al. (2009) study the case where instead of costs there are revenues for visiting a city. Toriello and Uhan (2013) study a version with asymmetric costs. Engevall et al. (1998, 2004) present applications of cost allocation in a TSP arising at an oil and gas company. Berger and Bierwirth (2010) study a network of collaborating carriers where the underlying transportation problem is a TSP.

2.3. Vehicle routing

The vehicle routing problem (VRP) consists of finding a set of vehicle routes through customer locations, so that each route starts and ends at a common location (see, e.g., a review by Laporte, 1992b). Usually a capacity restriction is imposed on the vehicles and each location has a specified demand quantity. A cost allocation problem linked to the VRP consists of splitting the cost of the routes among the visited cities. The first article dealing with the cost allocation problem in the VRP is Göthe-Lundgren et al. (1996). Later on, Engevall et al. (2004) study an application in the oil and gas industry. Krajewska and Kopfer (2006) and Krajewska et al. (2008) study cost allocation in a VRP arising in freight forwarding companies. Liu et al. (2010) perform numerical computations based on Krajewska et al. (2008). The VRP is also among several combinatorial problems where Caprara and Letchford (2010) address the cost allocation problem. Cruijssen et al. (2010) study cost allocation in a VRP with time windows. The version studied by Dahl and Derigs (2011) includes time windows and pickup and delivery in a dynamic setting. Also dynamic is the version studied in Zakharov and Shchegryaev (2015). The version in Vanovermeire and Sörensen (2014b) is stated as a capacitated multiperiod VRP. Özener (2014) studies a VRP involving emission costs.

2.4. Joint distribution

Several problems of collaborative transportation arise when two or more organizations collaborate through joint distribution. We have included in this class of problems a broad range of variations. Hamers et al. (1999) introduce a class of cooperative combinatorial optimization games called delivery games. These are associated to the Chinese postman problem, which consists of finding a route of minimal costs where all edges in a connected graph are visited at least once. Later work on cost allocation in this problem is developed by Granot et al. (2011).

Another problem of joint distribution originates when different shippers (companies, individuals) bundle their orders and procure transport services from a same carrier. The motivation here is that

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companies together can negotiate better rates with the carrier instead of negotiating individually, for example by requiring a full truckload instead of a less-than-truckload service. Then, the companies need to decide how to share the total cost of transport procurement. Cost allocation in this type of problems is addressed in Özener and Ergun (2008), Audy et al. (2011), Yilmaz and Savasaneril (2012), Lozano et al. (2013), Nguyen et al. (2014), Vanovermeire and Sörensen (2014a, 2014b), and Vanovermeire et al. (2014).

In providing transport services, the carriers can also collaborate by jointly optimizing the fulfillment of their delivery requirements. Instead of costs, the carriers here must usually solve a profit allocation problem. This has been addressed in Agarwal and Ergun (2010), Dai and Chen (2012), Dai and Chen (2015), and Yang et al. (2015). Hezarkhani et al. (2015) also study a carrier collaboration problem, but seen from the minimum cost perspective. The allocation problem here consists of how the carriers should split the savings obtained from collaboration.

2.5. Inventory

A fifth class of problems in this review includes transport operations involved in inventory problems. While the focus of these problems is on inventory (often by pooling resources), we include here only references where transportation plays an important role in the establishment of the collaboration. For example, Anupindi et al. (2001) study inventory pooling in decentralized distribution systems involving cooperative shipping and allocation decisions. Similar problems of inventory pooling with transshipments have been addressed in Wong et al. (2007) and Fang and Cho (2014).

Özener et al. (2013) address the cost allocation in an inventory routing problem, motivated in the context of an industrial gas company that operates a vendor-managed inventory resupply policy. Anily and Haviv (2007) address the cost allocation in an infinite-horizon deterministic single-warehouse joint replenishment model, where a number of retailers lease the reordering or transportation of their supplies, as well as their storage activities, to a third-party logistic provider. A similar variation appears in the inventory transportation system studied in Fiestras-Janeiro et al. (2012), where multiple agents place joint orders using an economic order quantity policy. The cost in this system can be written as the sum of two components, one due to common setup costs and other due to firm-dependent transportation. Additional work on this setting is presented in Fiestras-Janeiro et al. (2013).

Although not presented as an inventory problem, we have included in this class the work by Cheng et al. (2013), which in addition to transportation costs considers two types of typical inventory costs, namely holding costs and stockout costs. The problem here is to allocate the cost of empty railcar miles in a dynamic railroad network.

2.6. Related problems

Note in addition to the five classes of problems that constitute the focus of our review, there are other problems related to transportation and cost allocation. These include problems of infrastructure sharing, such as highways and bridges (Ghaeli et al., 2000; Lee and Garcia-Diaz, 2007). Other related problems arise in passenger transportation systems, where the question is how to allocate

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costs among different communities served by the system or to fix a fair tariff per passenger (see, e.g., Sánchez-Soriano et al., 2002). In the context of airlines, a typical cost allocation problem concerns the use of facilities at an airport, for which we refer the reader to Fiestras-Janeiro et al. (2011). Airlines, as well as shipping, have also motivated the study of strategic alliances, for which we refer the reader to Cruijssen et al. (2007). Network games, although not necessarily presented as transportation, may also be related to the types of problem included in this review (see, e.g., Rosenthal, 1991; Jackson, 2005; Quant et al., 2006).

3. Cost allocation

Most problems in collaborative transportation use concepts and methods from cooperative game theory. In this section, we summarize some of these concepts and the cost allocation methods used in the literature.

3.1. Cooperative game theory concepts

Let $N = \{1, \ldots, n\}$ be the set of all players or so called "grand coalition." Let K be the set of all subsets of N. The characteristic function $C : K \to \mathbb{R}$ assigns to each coalition S in K the *cost* of coalition S: This cost is usually calculated by solving an optimization problem for the corresponding coalition S. By convention, $C(\emptyset) = 0$. The pair (N, C) defines a transferable utility game.

A preimputation or cost allocation vector $x = (x_1, \ldots, x_n)$ assigns to each player j in N a quantity $x_j \in \mathbb{R}$ such that the following condition holds:

$$\sum_{j \in N} x_j = C(N). \tag{1}$$

Equation (1), so-called "efficiency" condition, states that the cost of the grand coalition N is split among its members according to the allocation x.

An allocation vector x satisfies "rationality," if the following constraints hold:

$$\sum_{j \in S} x_j \leq C(S) \, \forall S \in K. \tag{2}$$

Constraints (2) state that there is no subset S of players such that should they form a coalition separately from the rest they would perceive less total cost than the total cost allocated to them by x. In the case of S containing only one player, this corresponds to the so-called "individually rational condition," which states that the cost allocated to each player j must not be greater than its stand-alone cost $C(\{j\})$.

The core of the game is the set of preimputations that satisfy rationality, that is,

$$Core(C) = \{x \in \mathbb{R}^n : \sum_{j \in N} x_j = C(N), \ \sum_{j \in S} x_j \leq C(S) \, \forall S \in K\}. \tag{3}$$

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An allocation in the core provides stability, in the sense that there is no subset S such that its players would get a better outcome by deviating from the grand coalition N.

3.2. Allocation methods

We have found that more than 40 different cost allocation methods have been used in the literature on collaborative transportation. Several of these methods come from previous work on cooperative game theory, so we refer to them as "traditional methods." Many other methods have been proposed as result of the particular context of the problem studied in the articles, so we refer to them as "ad hoc methods." Table 3 gives an account of the frequency in which the methods appear in the literature of collaborative transportation. Note here the apparition of a method in an article does not necessarily mean it is the solution proposed by the authors, but it may also be used for benchmark purposes or to derive some properties.

3.2.1. Traditional methods

Due to the vast list of traditional cost allocation methods, we limit ourselves to give a brief definition of only those that appear in more than two articles.

Shapley value. The allocation by Shapley (1953) is one of the most used concepts in cooperative games. As can be seen in Table 3, it is also the traditional method with most apparitions in collaborative transportation, accounting for 23 articles. This method allocates to each player j an average of the marginal costs it implies when entering coalitions, as follows:

$$x_j = \sum_{S \subseteq N: j \in S} \left[\frac{(n - |S|)!(|S| - 1)!}{n!} \right] \cdot [C(S) - C(S \setminus \{j\})] \quad \forall j \in N. \tag{4}$$

The Shapley allocation is the only one satisfying four celebre axioms: efficiency, symmetry, dummy property and additivity. Another appealing feature of this allocation is that it is unique. However, in general, it does not belong to the core (it does so only if the game satisfies a convexity condition; Shapley, 1971).

Proportional methods. Proportional methods are those where each player j is assigned a share α_j of the total cost $C(N)$, that is

$$x_j = \alpha_j \cdot C(N) \quad \forall j \in N, \tag{5}$$

where $\sum_{j \in N} \alpha_j = 1$. The share values α_j can be defined according to different criteria. The simplest one is the "egalitarian" method (Dror, 1990; Berger and Bierwirth, 2010; Massol and Tchung-Ming, 2010; Lehoux et al., 2011), which assigns equal cost shares to all the players, that is, $x_j = C(N)/n \; \forall j \in N$. Some other frequent criteria to define the weights α_j are the demand quantities (Engevall et al., 2004; Wong et al., 2007; Frisk et al., 2010; Massol and Tchung-Ming, 2010; Nguyen et al., 2014; Flisberg et al., 2015) and the stand alone costs (Fishburn and Pollak, 1983; Özener and Ergun, 2008; Liu et al., 2010; Audy et al., 2012a; Özener et al., 2013; Özener, 2014; Sun et al., 2015).

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Table 3
Cost allocation methods used in literature on collaborative transportation

Method	No. of articles	References
Ad hoc	31	Potters et al. (1992), Derks and Kuipers (1997), Engevall et al. (1998), Faigle et al. (1998), Hamers et al. (1999), Engevall et al. (2004), Anily and Haviv (2007), Özener and Ergun (2008), Estévez-Fernández et al. (2009), Agarwal and Ergun (2010), Caprara and Letchford (2010), Frisk et al. (2010), Liu et al. (2010), Audy et al. (2011), Dahl and Derigs (2011), Audy et al. (2012a), Dai and Chen (2012), Fiestras-Janeiro et al. (2012), Yilmaz and Savasaneril (2012), Cheng et al. (2013), Fiestras-Janeiro et al. (2013), Özener et al. (2013), Toriello and Uhan (2013), Özener (2014), Vanovermeire and Sörensen (2014b), Dai and Chen (2015), Flisberg et al. (2015), Guajardo and Rönnqvist (2015), Hezarkhani et al. (2015), Sun et al. (2015), Yang et al. (2015).
Shapley	23	Dror (1990), Engevall et al. (1998), Engevall et al. (2004), Wong et al. (2007), Krajewska et al. (2008), Agarwal and Ergun (2010), Cruijssen et al. (2010), Frisk et al. (2010), Liu et al. (2010), Massol and Tchung-Ming (2010), Fiestras-Janeiro et al. (2012), Yengin (2012), Cheng et al. (2013), Lozano et al. (2013), Özener et al. (2013), Fang and Cho (2014), Vanovermeire et al. (2014), Vanovermeire and Sörensen (2014a), Guajardo et al. (2015), Hezarkhani et al. (2015), Sun et al. (2015), Wang et al. (2015), Zakharov and Shchegryaev (2015).
Proportional	18	Fishburn and Pollak (1983), Dror (1990), Engevall et al. (2004), Krajewska and Kopfer (2006), Wong et al. (2007), Özener and Ergun (2008), Berger and Bierwirth (2010), Frisk et al. (2010), Liu et al. (2010), Massol and Tchung-Ming (2010), Lehoux et al. (2011) Audy et al. (2012a), Özener et al. (2013), Nguyen et al. (2014), Özener (2014), Flisberg et al. (2015), Hezarkhani et al. (2015), Sun et al. (2015).
Nucleolus	12	Göthe-Lundgren et al. (1996), Engevall et al. (1998), Sakawa et al. (2001), Engevall et al. (2004), Agarwal and Ergun (2010), Frisk et al. (2010), Liu et al. (2010), Massol and Tchung-Ming (2010), Granot et al. (2011), Lozano et al. (2013), Guajardo et al. (2015), Hezarkhani et al. (2015).
Dual	8	Anupindi et al. (2001), Engevall et al. (2004), Bläser and Ram (2008), Agarwal and Ergun (2010), Frisk et al. (2010), Özener et al. (2013), Fang and Cho (2014), Özener (2014).
Marginal	8	Dror (1990),Engevall et al. (2004), Frisk et al. (2010), Massol and Tchung-Ming (2010), Dahl and Derigs (2011), Fang and Cho (2014), Flisberg et al. (2015), Hezarkhani et al. (2015).
τ-Value	2	Engevall et al. (1998), Lozano et al. (2013)
Aumann-Shapley	1	Samet et al. (1984)
Core centre	1	Lozano et al. (2013)
Minmax core	1	Lozano et al. (2013)
Disruption nucleolus	1	Massol and Tchung-Ming (2010)
Per capita nucleolus	1	Massol and Tchung-Ming (2010)
Modiclus	1	Guajardo et al. (2015)
SM-nucleolus	1	Guajardo et al. (2015)

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Nucleolus. Originally defined by Schmeidler (1969) in the form of a profit game, we present its definition adapted to a cost sharing game (N, C). Define the excess of coalition S at x in game (N, C) as $\varepsilon(x, C, S) = C(S) - \sum_{j \in S} x_j$. The excess of a coalition S at an allocation x can be interpreted as a measure of satisfaction of the coalition with this allocation. The larger the excess of S, the more satisfied coalition S is, in the sense that it achieves larger savings. For a game (N, C), define the excess vector at x as $e(x, C) = (\varepsilon(x, C, S_1), \ldots, \varepsilon(x, C, S_p))$, where the sets S_i represent the coalitions in $K \setminus \{N\}$ and $p = 2^n - 1$. An imputation is a preimputation that satisfies the individual rationality condition, that is, $x_j \leq C(\{j\}) \, \forall j \in N$. Denote the set of imputations as $\mathcal{X}$. A vector $y \in \mathbb{R}^p$ is lexicographically greater than $\bar{y}$ (written $y \succeq \bar{y}$) if either $y = \bar{y}$ or there exists $h \in \{1, \ldots, p\}$ such that $y_h > \bar{y}_h$ and $y_i = \bar{y}_i \, \forall i < h$. The nucleolus $\mathcal{N}$ of the cost sharing game (N, C) can be defined as

$$\mathcal{N} = \{x \in \mathcal{X} : \theta(e(x, C)) \succeq \theta(e(y, C)) \, \forall y \in \mathcal{X}\}, \tag{6}$$

where $\theta(e(x, C))$ is the vector that results from arranging the components of the excess vector in nondecreasing order. Thus, the nucleolus is the set of imputations that lexicographically maximizes the excess vector. Appealing well-known features of the nucleolus is that it is unique and that it always belongs to the core whenever this is nonempty. The computation of the nucleolus is somewhat more complex than formulae-based methods. Due to this reason, its use is not rarely mistaken (Guajardo and Jörnsten, 2015). Several algorithms have been proposed, usually based on a sequence of linear programs, as in Kopelowitz (1967) and Fromen (1997).

Methods based on dual or shadow prices. In transportation problems, the cost function is usually obtained by solving a linear programming model. The dual program associated to this underlying linear programming model can be used to generate an allocation, since its optimal objective value is the same as the primal model. The allocation to each player can be calculated by computing its contribution to the dual objective function value. Our list of articles accounts for eight apparitions of methods based on dual prices, as can be seen in Table 3.

Methods based on marginal or separable and nonseparable costs. The marginal or separable cost of player j is $m_j = C(N) - C(N \setminus \{j\})$. By subtracting all the separable costs from the grand coalition cost, the nonseparable cost $g(N) = C(N) - \sum_{j \in N} m_j$ is obtained. Engevall et al. (2004) and Massol and Tchung-Ming (2010) use the marginal cost m_j as allocation, but note that in general they do not lead to efficiency. Dahl and Derigs (2011) use the marginal cost concept under several variations and Fang and Cho (2014) use a marginal cost method proposed by Jackson and Wolinsky (1996). Another method based on marginal costs, widely known for its use in allocating costs of multipurpose water reservoirs (Straffin and Heaney, 1981), is the alternative cost avoided method (ACA). This method allocates to player j a cost $u_j = m_j + [r_j / (\sum_{i \in N} r_i)] \cdot g(N)$, where $r_j = C(\{j\}) - m_j$ is the so-called alternate cost avoided of player j. In collaborative transportation, ACA appears in Frisk et al. (2010), Massol and Tchung-Ming (2010), Flisberg et al. (2015), and Hezarkhani et al. (2015). The equal charge method (ECM) is similar to ACA, but the nonseparable cost is divided equally among the players, that is, $u_j = m_j + g(N)/n$. ECM appears in Dror (1990), Frisk et al. (2010), and Flisberg et al. (2015). The cost gap method (CGM), on its hand, divides the

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nonseparable cost in proportion to $w_j = \min_{S:j\in S} C(S) - \sum_{j\in S} m_j$. CGM appears in Dror (1990) and Frisk et al. (2010).

3.2.2. Ad hoc methods

The literature in collaborative transportation has originated many new cost allocation methods. As reported in Table 3, our review accounts for 31 articles including ad hoc methods. A detailed specification of these methods would require a number of definitions concerning the particular context where they have been motivated. Instead, we provide a brief outline of these methods in Table 4 and refer the reader to the references therein for a detailed definition. Note several of these methods are modified versions of the traditional methods. Another frequent approach in these methods is finding core allocations by solving a mathematical programming model where the efficiency and rationality conditions are explicitly incorporated as constraints and the objective function captures some fairness measure.

4. Other characteristics

4.1. Data features

Some data features are presented in Table 5 for all articles including numerical computations. These numerical computations range from illustrative examples to thorough case studies. The second column of Table 5 shows the number of players n involved in the computations. This number varies between 2 and 80. Note the dimension of the cases using industrial real data is, in general, considerably higher than the dimension involved in most numerical computations without real industrial data. For articles using industrial data, we have also added a column with the potential savings from collaboration, if reported. It can be seen that collaboration usually renders significant benefits, ranging from 5.99% to 46% cost savings.

4.2. Sector

A large part of the articles included in this review are motivated in the context of carrier companies providing ground transportation services. The list here includes Krajewska and Kopfer (2006), Krajewska et al. (2008), Özener and Ergun (2008), Agarwal and Ergun (2010), Berger and Bierwirth (2010), Liu et al. (2010), Dahl and Derigs (2011), Dai and Chen (2012), Özener (2014), Vanovermeire and Sörensen (2014b), Dai and Chen (2015), Yang et al. (2015), and Zakharov and Shchegryaev (2015). Others arise in the context of railcar companies (Cheng et al., 2013) and the maritime transportation sector (Yilmaz and Savasaneril, 2012). Logistics service providers (Vanovermeire and Sörensen, 2014a; Hezarkhani et al., 2015), retailers (Anupindi et al., 2001; Cruijssen et al., 2010), or a combination of these two (Anily and Haviv, 2007; Vanovermeire et al., 2014) also provide contexts for collaborative transportation.

Another protagonist in the literature is the natural resources sector. This sector is represented by articles in agriculture (Nguyen et al., 2014), forestry (Frisk et al., 2010; Lehoux et al., 2011; Audy

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Table 4
Ad hoc cost allocation methods in the literature on collaborative transportation

References	Method or principle
Potters et al. (1992)	A share of the stand-alone cost plus a share of the marginal cost.
Derks and Kuipers (1997)	Recursive definition for which a game theoretic interpretation is given.
Engevall et al. (1998, 2004)	(a) Demand nucleolus: nucleolus modified by multiplying the excess by demand; (b) "The principle implied by Norsk Hydro": an allocation based on cost per demand.
Faigle et al. (1998)	LP-based allocation rule based on moat packing and a geometric argument.
Hamers et al. (1999)	γ method: each player pays an equal share of the travel cost incurred over a bridge needed to visit him.
Anily and Haviv (2007)	A core allocation where all retailers pay their own minor setup costs and holding costs, and the major setup cost is paid by some of the retailers.
Özener and Ergun (2008)	(a) Minimum range method: core allocation as close as possible to the original lane costs; (b) methods with minimum liability restriction with relaxed efficiency or with a minimum percent deviation from stability.
Estévez-Fernández et al. (2009)	Recursive definition based on Derks and Kuipers (1997) and connected coalitions.
Agarwal and Ergun (2010)	Based on inverse optimization, the profit allocation to a carrier is the solution to its single carrier problem plus a side payment share.
Caprara and Letchford (2010)	Three methods based on column generation, row generation and a combination of both.
Frisk et al. (2010),	Equal profit method (EPM): a core stable allocation such that the maximum difference in pairwise relative savings is minimized.
Liu et al. (2010)	Based on a weighted relative savings model.
Audy et al. (2011)	(a) EPM modified: assures a minimum percentage of savings and introduces three nontransferrable costs for each company; (b) ACA modified:
Dahl and Derigs (2011)	Based on practice of a company, it distinguishes whether the order is served by a dedicated tour or inserted into an already existing tour.
Audy et al. (2012a)	Modification of the proportional to stand-alone cost according to different versions of leading company behavior.
Dai and Chen (2012)	(a) Core guaranteed closest to Shapley; (b) core guaranteed proportional based on Kalai (1977); (c) core guaranteed contribution-based: introduces a measure of the contribution of each carrier and minimizes the difference between the contribution-based allocation ratios of any two carriers.
Fiestras-Janeiro et al. (2012)	Line rule: inspired in Shapley value, but requires less computational effort and relates better with the core.
Yilmaz and Savasaneril (2012)	Allocation that maximizes the deviation from the minimum expected payment.
Cheng et al. (2013)	Shapley modified: mathematical programming model that minimizes deviations from Shapley value.
Fiestras-Janeiro et al. (2013)	Average of the Marginal vectors with an Extreme agent First (AMEF).
Özener et al. (2013)	(a) Allocation from Faigle et al. (1998) adapted to inventory routing; (b) method used in practice that slightly modifies the proportional method, based on a mileage cost with customer's usage rate and a tank size factor.
Toriello and Uhan (2013)	Based on the parsimonious property and a connection to linear production games.
Özener (2014)	Shapley modified, such that only some characteristic function values are needed.
Vanovermeire and Sörensen (2014b)	Modified versions of Shapley, nucleolus and ECM. The modification incorporates the relevance of a member to become flexible (e.g., in terms of delivery times).
Dai and Chen (2015)	Proportional Egalitarian Core (PEC), defined by a model which minimizes the weighted sum of the square profits of all players.

Continued

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Table 4
Continued

References	Method or principle
Flisberg et al. (2015)	Semicore modifications of the EPM and nucleolus.
Guajardo and Rönnqvist (2015)	Based on a mixed integer programming model that simultaneously defines an allocation and a coalition structure.
Hezarkhani et al. (2015)	(a) ϵ-Stable allocation closest to the allocation proportional to stand-alone cost. (b) Also uses the EPM as benchmark.
Sun et al. (2015)	(a) Allocation from Faigle et al. (1998). (b) Contribution Constrained Packing Model allocation.
Yang et al. (2015)	A transfer payment contract including fixed and variable fees based on a Nash bargaining model.

et al., 2012a; Flisberg et al., 2015; Guajardo and Rönnqvist, 2015), and the oil and gas industry (Engevall et al., 1998, 2004; Massol and Tchung-Ming, 2010; Özener et al., 2013; Guajardo and Rönnqvist, 2015). Other real contexts where the collaborative transportation has been studied include cases in the furniture industry (Audy et al., 2011) and a housing material manufacturer (Sakawa et al., 2001).

4.3. Performance

Given the great variety of cost allocation methods, a natural concern is discerning whether a method is "better" than another one. In general, there is no closed answer to this question and a proper answer would depend on the performance measures and notion of fairness defined for a particular problem.

4.3.1. Stability and comparison measures

A main criterion required to the allocation methods is satisfying some notion of "stability." In this, the core concept is one of the most frequent ones in the literature, including Tamir (1989), Dror (1990), Potters et al. (1992), Göthe-Lundgren et al. (1996), Derks and Kuipers (1997), Engevall et al. (1998), Hamers et al. (1999), Anupindi et al. (2001), Sakawa et al. (2001), Engevall et al. (2004), Anily and Haviv (2007), Wong et al. (2007), Krajewska et al. (2008), Özener and Ergun (2008), Estévez-Fernández et al. (2009), Agarwal and Ergun (2010), Caprara and Letchford (2010), Frisk et al. (2010), Massol and Tchung-Ming (2010), Audy et al. (2011), Granot et al. (2011), Dai and Chen (2012), Fiestras-Janeiro et al. (2012), Yengin (2012), Cheng et al. (2013), Fiestras-Janeiro et al. (2013), Lozano et al. (2013), Fang and Cho (2014), Nguyen et al. (2014), Vanovermeire and Sörensen (2014b), and Dai and Chen (2015).

Other relaxed versions of the core have also been studied in collaborative transportation. These include the ϵ-approximation core, which relaxes the rationality constraints (2) by an ϵ value, either as a fraction (Faigle et al., 1998; Bläser and Ram, 2008) or as an absolute term (Hezarkhani et al., 2015). Another version is the α-core, which relaxes the efficiency constraint (1) by some fraction (Toriello and Uhan, 2013). Another relaxed version called the "semicore" is studied by Flisberg

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Table 5
Data features of the articles presenting numerical computations

Reference	n	Industrial data	Savings
Özener et al. (2013)	25, 26, 50, 70, 80	√	
Dahl and Derigs (2011)	50	√	13.85%
Özener (2014)	25, 50		
Flisberg et al. (2015)	29	√	5.99%, 22.18%
Göthe-Lundgren et al. (1996)	6, 25		
Engevall et al. (2004)	21	√	46%
Guajardo and Rönnqvist (2015)	8, 16, 17	√	8.6%, 9.3%, 14.2%
Sun et al. (2015)	14		
Massol and Tchung-Ming (2010)	11, 12	√	10.5%, 11.9%, 12.9%
Özener and Ergun (2008)	2, 5, 10	√	
Lozano et al. (2013)	4, 8		
Frisk et al. (2010)	3, 8	√	8.6%, 9.3%, 14.2%
Audy et al. (2012a)	3, 8	√	8.6%, 9.3%, 14.2%
Guajardo et al. (2015)	3, 8	√	8.6%, 9.3%, 14.2%
Engevall et al. (1998)	5	√	
Hamers et al. (1999)	5		
Sakawa et al. (2001)	5	√	
Nguyen et al. (2014)	5	√	
Dai and Chen (2015)	3, 5		
Krajewska et al. (2008)	2, 5	√	11.50%
Potters et al. (1992)	4		
Fiestras-Janeiro et al. (2013)	4		
Zakharov and Shchegryaev (2015)	4		
Agarwal and Ergun (2010)	2, 3, 4		
Cruijssen et al. (2010)	3, 4	√	23.70%
Audy et al. (2011)	3, 4	√	12.90%
Fang and Cho (2014)	3, 4		
Wang et al. (2015)	3, 4		
Krajewska and Kopfer (2006)	3		
Estévez-Fernández et al. (2009)	3		
Berger and Bierwirth (2010)	3		
Liu et al. (2010)	3		
Dai and Chen (2012)	3		
Fiestras-Janeiro et al. (2012)	3		
Yengin (2012)	3		
Cheng et al. (2013)	3		
Toriello and Uhan (2013)	3		
Vanovermeire et al. (2014)	3	√	25.83%, 41.81%
Vanovermeire and Sörensen (2014a)	3		
Hezarkhani et al. (2015)	3		
Fishburn and Pollak (1983)	2, 3		
Anupindi et al. (2001)	2, 3		
Wong et al. (2007)	2, 3		
Samet et al. (1984)	2		
Anily and Haviv (2007)	2		
Lehoux et al. (2011)	2	√	4%
Vanovermeire and Sörensen (2014b)	2		
Yang et al. (2015)	2		

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et al. (2015), which includes the rationality constraints only for the singleton coalitions of the type $S = \{j\}$ and its complement $S = N \setminus \{j\}$ for all $j \in N$. Although the semicore is a weaker concept than the core in terms of stability, it is more practical, as it requires calculating the characteristic function values for only $2n + 1$ coalitions instead of the $2^n - 1$ required by the core.

An "approximate stability assessment" procedure is developed in Özener et al. (2013), based on sampling of subsets and testing the stability of coalitions on these sampled subsets only. The subsets are sampled according to geographic proximity, pursuing to select coalitions where it is likely to find high synergy among its members. Özener (2014) reports this procedure proves to be very effective and results in a significant decrease in computational effort.

Another problem linked to the stability of cost allocations refers to which coalitions should form and the sequence in which they should form. Cruijssen et al. (2010) and Wang et al. (2015) have studied strictly monotonic paths for this problem. Audy et al. (2012a) study paths starting from a leader player and uses the solution concept of subgame perfect Nash equilibrium to identify which path will realize. The Nash equilibrium concept is also studied in a competitive version of the game in Wong et al. (2007). Guajardo and Rönnqvist (2015) approach the cost allocation simultaneously with a coalition structure problem, where the core concept includes not only internal stability within the players belonging to a same coalition in the structure but also with respect to players in other coalitions. Guajardo et al. (2015) study this problem using several cost allocation methods as prescribed rules and verify whether the rationality constraints hold or are violated.

In addition to the stability criteria mentioned above, the comparison of the methods is usually linked to the measure of fairness adopted in the problem. As we reported in Table 4, there are many ad hoc methods in the literature and often they are motivated by a particular measure of fairness. As expected, it is usually the case that the ad hoc methods perform better according to the particular measure that motivate them. For example, the notion of fairness in Frisk et al. (2010) is trying to find an allocation in the core such that the savings of the companies relative to their stand-alone costs are as similar as possible. The EPM allocation is constructed specifically to achieve this goal, by a linear programming model, where the objective function minimizes the maximum difference among pairwise relative savings. In consequence, the results obtained by the EPM outperform all the other methods in this respect. Lozano et al. (2013) compare methods according to the Mean Absolute Deviation between the cost savings shares allocated to the different companies.

4.3.2. Findings

We conclude this section by highlighting some of the findings in the literature of collaborative transportation. The Shapley allocation lies in the core of the numerical instances tested in Krajewska et al. (2008). Vanovermeire et al. (2014) point out that the Shapley value gives incentives to be flexible and collaborative. It is well known, however, that in general Shapley values do not necessarily belong to the core (for an example in transportation planning, see Guajardo et al., 2015). In Vanovermeire and Sörensen (2014b), the nucleolus and ECM reward the flexible partner with the highest impact to a greater extent than the Shapley value.

The nucleolus always lies in the core when the core is nonempty. It has, however, motivated some criticism because it does not satisfy some desirable properties, such as monotonicity in the

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aggregate required in Massol and Tchung-Ming (2010). They overcome this by using the per capita nucleolus variant. Engevall et al. (1998) introduce the demand nucleolus and find that the importance of coalitions having a large demand is reduced in comparison to the nucleolus. Other alternative concepts derived from the nucleolus are the modiclus (introduced by Sudhölter, 1997 as the "modified nucleolus") and the SM-nucleolus (Tarashnina, 2011). These two methods take into account what is called the blocking power. Guajardo et al. (2015) empirically investigate these methods in a case in forest transportation and find that players more centrally located tend to benefit from the nucleolus, which allocates less cost to them than the modiclus or the SM-nucleolus. The modiclus and the SM-nucleolus tend to correct the relative importance of these players with respect to those in more peripheral areas. The blocking power acknowledges that these more peripheral companies, as a block, are still crucial to the collaboration, despite among themselves they have little opportunities for collaboration.

Proportional cost allocation methods are easy to communicate to practitioners. However, except for their positive performance in Nguyen et al. (2014), the literature presents evidence of weakness of these methods. For example, Özener and Ergun (2008) find that for proportional cost allocations the percentage of deviation from stability is approximately 25% over all tested instances, which suggests that there exists a significant risk for the disintegration of the collaboration. Özener et al. (2013) assert that proportional allocation schemes typically used in practice ignore the synergies among customers. In their empirical analysis, they find that a proportional cost allocation method has a high number of instable subsets and extremely high maximum instability values. The average of maximum percentage instability value is around 35%, and the maximum of this value is close to 50%. In the same analysis, the Shapley value approximation performs only slightly better than the proportional cost allocation method, while the duality-based cost allocation methods perform significantly better than the other methods. In Özener (2014), the duality-based method provides significantly better allocations especially compared to the proportional and individual solution based allocation methods (this superior performance is even more significant in larger instances). Structural results on core stability of a dual price allocation are presented in Anupindi et al. (2001). Fang and Cho (2014) point out, however, that this dual-price allocation may not be intuitively appealing. Moreover, in the transportation planning problem addressed in Frisk et al. (2010), the numerical results obtained for the dual-price allocation does not belong to the core. The proportional allocation and the ECM in this case does not belong to the core either, while the Shapley allocation and some allocations based on marginal costs (ACA and CGM) belong to the core. Numerical findings in Flisberg et al. (2015) reveal that allocations based on proportional methods and the ECM do not even belong to the semicore, while the allocations obtained by ACA and the adapted versions of the EPM and the nucleolus belong to the semicore.

Overall, we observe that the findings in the literature are usually result of intensive numerical experimentation and, despite some traditional methods repetitively appear in the literature, there is no broad consensus on their acceptance.

5. Concluding remarks

Different forms of collaboration are becoming more important for the competitiveness of companies. These range from strategic alliances to operative coordination, and cover both vertical and

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horizontal collaboration. In this article, we survey the recent work on cost allocation models in a selected but important part of collaborative logistics. These are transportation planning, traveling salesman, vehicle routing, distribution, and inventory problems. The selected areas include well-known problems where many models and methods have been used. Some of these methods come from previous literature, often from cooperative game theory, and others are ad hoc methods. The basis may be well founded but the ad hoc versions are sometimes not well motivated from a practice point of view. This survey provides a good basis for researchers both in what is done and also in the description of the main approaches suggested for those not familiar with the different approaches.

The survey also describes the main cost allocation methods found in the literature. While the literature on collaborative transportation has succeeded in adopting methods from game theory and in introducing new methods, the task of understanding the differences in the cost allocations obtained by such different methods is less accomplished. Often the new methods obey a particular notion or measure of fairness and, in consequence, they are expected to perform better with respect to the measures they are designed for. It is clear that many proposed allocation models are not based on practical requirements from actual contexts of transportation and rather theoretical combinations of already defined allocation models.

The area is expanding rapidly and there are many interesting research opportunities. One is on the construction and management of the collaboration. A very important question is how to describe and measure the performance for different stakeholder in the collaboration. Another question is how to develop a process for different stakeholders to agree on a sharing mechanism and have the incentives to stay in it. If the requirements from different players are somewhat different, it is interesting to study principles in how the additional requirements should impact the basic cost allocation. The building of a coalition is dependent on the strategy of the leader and this needs further research. Collaboration may involve sensitive information and some participants should not have access to other participants data. In such cases, it is important that the planning is done by an independent third-party logistics provider. In some cases, the same is also operating the plan and there are no problem. If the plans however are operated by the participants, it is important to make sure that sensitive information are not transferred through the plans. In the case a leader is also responsible for planning, the problem with sensitive information becomes more difficult and techniques and systems to overcome this need further research.

In many practical cases, there is a dynamical nature of the problems often overlooked in the literature even when the characteristic function is result of optimizing a model accounting for several time periods. The contribution to the coalition may change over time, which may also involve uncertainty in the outcomes. Hence, it is interesting how cost allocation can be based on dynamic games, contract setting or systems that allows for a dynamic change of the allocations in real time. Efforts in these direction include Dahl and Derigs (2011) and Furuhata et al. (2015).

Some problems are very large in terms of participants and the computational time for standard game theoretical models becomes too limiting due to the exponential growth in the number of coalitions with respect to the number of players. It is hence interesting to develop approximation approaches. Here, the approximation can be performed, for example, in computing the characteristic function values. Another approach could be based on approximating the description of the game, by omitting the computation of the characteristic function for some coalitions. In the same line, the cost allocation methods themselves could be based on approximations. Since it is often the case

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that the characteristic function value is the optimal objective value of an optimization model (linear programming, integer programming), operations research methods can play a fundamental role in approaching large-scale problems.

Finally, collaborative transportation may also be introduced in teaching programmes. Some pioneer efforts in this line are given by D'Amours and Rönnqvist (2013).

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